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Course Code: CSE205R02

Semester: IV

COMPUTER ARCHITECTURE

Course Objectives

This course empowers learners to grasp the fundamental components of ARM7 architecture, extending their knowledge to delve into the STM32 microcontroller for embedded system design. Through comprehensive instruction, learners will explore a wide array of architectural designs, with a particular focus on comprehending concurrency at multiple levels including instruction, thread, and data level parallelism.

UNIT - I

15 Periods

ARM Architecture and Instruction Set: RISC Design Philosophy - ARM Design Philosophy - Embedded System Hardware - Embedded System Software – ARM Data Flow Model - Registers - Current Program Status Register - Pipeline - Exceptions - Interrupts and the Vector Table - ARM Instruction Set - Thumb Instruction Set.

UNIT - II

15 Periods

STM32: ARM Cortex-M3 based Microcontroller Architecture: History of ARM and Microcontrollers - I/O Port – UART Serial Port - Timer – Interrupt and Exception – ADC - I2C : Pins, SFRs and Application Programming.

UNIT - III

15 Periods

Advanced Architecture Features: Instruction-Level Parallelism: Concepts and Challenges-Basic Compiler Techniques for Exposing ILP-Reducing Branch Costs with Advanced Branch Prediction-Overcoming Data Hazards with Dynamic Scheduling-Dynamic Scheduling: Examples and the Algorithm-Hardware-Based Speculation

UNIT - IV

15 Periods

Vector Architecture and Parallelism: Data level parallelism: Introduction - Vector Architecture - SIMD Instruction Set Extensions for Multimedia - Graphics Processing Units - Detecting and Enhancing Loop - Level Parallelism - **Thread-Level Parallelism:** Introduction - Centralized Shared-Memory Architectures - Distributed Shared-Memory and Directory-Based Coherence - Synchronization.

TEXT BOOKS

1. Andrew N Sloss, Dominic Symes and Chris Wright, "ARM System Developers Guide: Designing and optimizing system software", Morgan Kaufmann, 2004.
2. Muhammad Ali Mazidi, Sarmad Naimi, SepehrNaimi, "The stm32f103 arm microcontroller and embedded systems: Using assembly and c", First Edition, MicroDigitalEd, 2020.
3. John L. Hennessy, David A. Patterson. *Computer Architecture - A Quantitative Approach*, Morgan Kaufmann is an imprint of Elsevier, Sixth Edition, 2019.

REFERENCES

1. William Stallings. *Computer Organization and Architecture - Designing for Performance*, Pearson Education, Tenth edition, 2006.

ONLINE MATERIALS

1. https://onlinecourses.nptel.ac.in/noc21_cs47/preview
2. <https://nptel.ac.in/courses/106/105/106105193/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Describe the architecture, instruction set and advanced features of ARM7 processor
Unit II	• Identify the significance of STM32 controller along with its Pin and register description with its features for designing an embedded system
Unit III	• Select techniques for extending basic pipelining to increase parallelism explored at instruction level
Unit IV	• Illustrate vector architecture and GPU architecture • Extend enhancement of loop level parallelism and Thread Level Parallelism

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

CO No.	Course outcomes	HCL
CO1	Describe the architecture, instruction set and advanced features of ARM7 processor	K2
CO2	Identify the significance of ARM7 based Microcontroller along with its Pin and register description with its features for designing an Embedded / IoT system	K2
CO3	Ability to write C code to interface on-chip peripheral devices with STM32 microcontroller	K3
CO4	Select techniques for extending basic pipelining to increase parallelism explored at instruction level	K6
CO5	Illustrate vector architecture and GPU architecture to support Data Level parallelism	K2
CO6	Exploit Thread Level Parallelism through MIMD architecture	K6

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Course Code: CSE206R02
Semester: IV

COMPUTER SYSTEM DESIGN LABORATORY

Course Objectives

This course will help the learner to understand assembly language, I/O operations, Timer, ADC operations and Interrupts of ARM based Microcontroller architecture.

LIST OF EXPERIMENTS

Assembly Language Programming

1. Programs on Data transfer Instructions
2. Programs on ALU instructions
3. Programs on Stack

Interfacing On and Off Chip Peripherals with ARM

4. Programming GPIO
5. Designing Program flow control using Internal Timers
6. Implementation of Hardware Interrupt
7. Data transfer using Serial Communication
8. Interfacing Sensor using ADC
9. Implementation of I2C Communication
10. Interfacing Display Devices

Using GEM5

11. Program on Task context maintenance
12. Performance analysis of cache memory

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Develop Assembly language program for any given problem using ARM processors
- Demonstrate I/O operations for Advanced Microcontroller
- Demonstrate Timer operations
- Illustrate the ADC and UART functions in ARM board
- Demonstrate interrupt routines
- Analysis of system architecture using GEM5